

Location-Based Artifact Installation Interaction

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Abstract

Our project establishes digital 3D models of artifacts through digital scanning technology and develops proportionate interactive devices that can interact with projection images using spatial location technology. The project aims to commemorate the Southward Evacuation of the artifacts led by the Palace Museum in China during World War II. By allowing visitors to physically touch the artifact installations for experiential purposes, it breaks the traditional exhibition limitation of only being able to view artifacts from a distance. This approach enables audiences to gain a more comprehensive understanding of cultural heritage through a multi-sensory experience, thereby encouraging reflection on the value and significance of artifact preservation.

CCS Concepts

• Human-centered computing; • Interaction design; • Systems and tools for interaction design;

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1 Introduction

The Southward Evacuation of Artifacts was a wartime preservation movement organized by the Palace Museum in China. It stands as a historic miracle of cultural heritage protection, being the largest, most extensive, longest lasting, and most impactful effort of its kind during the global struggle against fascism. While China suffered significant losses of cultural relics during World War II, the artifacts from the Palace Museum were largely preserved with minimal loss. Over 19,000 crates of artifacts were safeguarded through more than two decades of difficult re-locations, ultimately ensuring their preservation in near-complete condition.

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Figure 1: Interactive exhibition of Southward Evacuation

In exhibition design, there have been numerous attempts to break the "no-touch policy" of museums, such as the use of interactive display cases [Pattakos et al., 2023]. Our project leverages hands-on interactive experiences to help visitors establish a deeper connection with artifacts. Interactive design can effectively stimulate visitors' curiosity, interest, and engagement [Roberts et al., 2018]. Our project (Figure 1) is based on the "Emperor Qianlong: One or Two" painting, allowing visitors to experience the scene of Emperor Qianlong appreciating ancient artifacts through the interaction of projected images and artifact installations. Six artifacts were selected from the painting, four of which underwent the Southward Evacuation, while the other two remained in Beijing without being relocated. The tumultuous fate of these artifacts reflects the harsh realities of Second World War and the determination of the Chinese people to protect cultural heritage.

When visitors pick up a interactive installation, it triggers the synchronized wall projection, displaying a digital 3D image corresponding to the artifact. The interactive installation, equipped with a positioning system, achieves real-time synchronization with the digital projection, allowing visitors to freely explore and appreciate the artifact from any angle. At the same time, the interactive experience is further enhanced by floor projections that display texture animations, showcasing the exquisite beauty of the artifacts, while the wall on back side project historical narratives and routes related to the specific Southward Evacuation movement.

2 System Workflow

On the hardware front, we engineered positioning chips and location system for the artifact models. After the digital scanning of the artifacts, we utilized 3D printing technology to produce tangible, interactive replicas [Dean, 2003]. On the software side, we developed an interactive system using Unity that enables visitors to engage with the artifacts in a highly immersive manner.

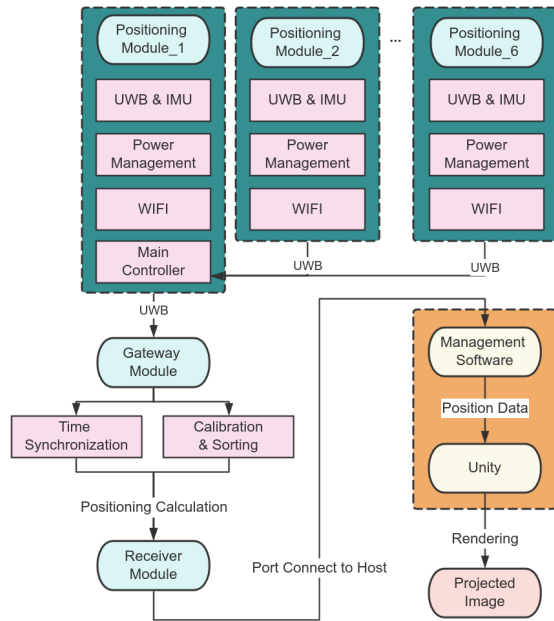


Figure 2: Workflow of the Location System

The operational workflow of the location system (Figure 2), focusing on the interaction between six positioning hardware modules and the real-time control of software operations. The system comprises six distinct Positioning Modules, each equipped with Ultra Wide-Band (UWB) and Inertial Measurement Unit (IMU) sensors, power management units, and Wi-Fi communication capabilities. Among these, Positioning Module 1 serves as the primary controller, linking with the other five modules and transmitting consolidated positioning data to the Gateway Module.

The Gateway Module acts as a central node, receiving UWB signals from all the positioning modules. It undertakes critical functions such as time synchronization and calibration & sorting to ensure the accuracy and reliability of the collected positioning data. Subsequently, the module performs positioning calculation before forwarding the processed data to the Receiver Module, which manages the connection to the host through a designated port.

On the software side, the Management Software processes the incoming position data, ensuring it is accurately reflected within the Unity engine. Unity then utilizes this data to accurately render the corresponding digital 3D models and associated content. The final output is a Projected Image that dynamically responds to the real-time movement of the physical artifacts.

This integration of hardware and software facilitates the interactive experience, where the positioning data from the six installations directly influences the visual content displayed on the walls, enabling users to engage with the artifact in an immersive manner (Figure 3). At the same time, the latency can be controlled within 75ms, enabling synchronized interaction.



Figure 3: Hardware and software display effects

3 Discussion

The project breaks through the traditional exhibition limitation of untouchable artifacts, enabling visitors to establish a deeper emotional connection with the artifacts through a multi-sensory experience. This approach allows for a more vivid understanding of the cultural essence embedded in the artifacts and encourages reflection on the importance and significance of artifact preservation.

One of the primary challenges faced in this project was maintaining precise synchronization between the physical artifacts and their digital counterparts. Variations in the latency of positioning data, especially in environments with potential interference, sometimes led to a slight desynchronization between the movements of the installation and the projected image. Additionally, the system's reliance on UWB and IMU technologies, while generally effective, presented occasional issues in complex environments where signal obstruction or reflection occurred, affecting the accuracy of the tracking.

To address these challenges, future optimizations will focus on improving the robustness of the location system. This includes exploring alternative or supplementary technologies, such as integrating computer vision-based tracking to enhance precision and reduce latency. We also plan to enhance the interactive experience by expanding the range of multi-sensory feedback, potentially incorporating haptic feedback to further bridge the gap between the physical and digital worlds. These improvements aim to provide visitors with a novel, immersive, and multi-sensory interactive XR experience.

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